

Incident Report: Zero Trust SOC Lab

This report documents a simulated cyber attack within a controlled Zero Trust environment monitored using Splunk SIEM.

Incident Summary

A simulated reconnaissance and scanning attack was launched from a Kali Linux attacker machine targeting internal network assets. The activity was detected and blocked by pfSense firewall rules configured under a Zero Trust model.

Environment

Attacker: Kali Linux Firewall: pfSense SIEM: Splunk Enterprise Target Network: LAN (192.168.1.0/24) DMZ Network: 192.168.2.0/24

Attack Details

Technique: Network Scanning (Nmap SYN Scan) Command Used: `nmap -Pn -sS 192.168.1.0/24`
MITRE ATT&CK: T1595 (Active Scanning)

Detection

Splunk SIEM detected repeated blocked traffic events using the query: `index=main "UFW BLOCK" | stats count by SRC`

Evidence

Firewall logs confirmed multiple blocked attempts from source IP 192.168.2.100 targeting various internal hosts and ports.

Impact

No successful intrusion occurred. All malicious traffic was blocked by Zero Trust firewall policies.

Response

Firewall rules enforced segmentation (DMZ to LAN blocked). Logs were collected and analyzed in Splunk. Alerts and dashboards were used for monitoring.

Conclusion

The Zero Trust architecture effectively prevented lateral movement and unauthorized access. The SOC workflow successfully detected and analyzed the attack.